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Back to the basics: mortality and fertility interactions during the demographic transition

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Mortality change has traditionally been considered a necessary prerequisite to any long-term reduction in fertility. In Frank Notestein's original statement of the demographic transition theory, societies went from high to low levels of fertility and mortality by means of a path in which mortality declined first and was followed at a shorter or longer lag by fertility.[1] Subsequently Abdel Omran outlined a number of scenarios of demographic transition, all of which reflected simultaneous or prior declines of mortality as a major factor stimulating fertility reduction.[2] While Omran's view of the epidemiological transition goes on to emphasize many other aspects of mortality decline, his view of it as an agent for fertility decline is similar to Notestein's.[3]

This understanding of the mortality–fertility synergy is based on two very general conceptions of historical reality which both Omran and Notestein share with many people working in the field of historical demography. For one, there has never been a society known to historians in which fertility was high and mortality low, or mortality high and fertility low, except in moments of epidemic or economic or political stress, for relatively brief periods of time, or among very specific historic sub-populations.[4] Normally mortality and fertility have ended up adjusting to each other more or less efficiently so as to ensure moderate rates of population growth. Moreover, high levels of fertility and mortality invariably were replaced by moderate ones during the demographic transition. Secondly, one of the reasons that the demographic transition holds such an allure for most historians is that it was a

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period of history in which societal traditions appear to have succumbed to the weight of rational choices made by families in order to ensure their best welfare in the face of changing circumstances. The initial instigator of the change, one would reasonably suspect, had to have been a decline in mortality with increasing numbers of children surviving beyond infancy. Faced with this reality, families made the decision to give individual choice priority over social norms and ended up curtailing their fertility. The entire context of social, economic and cultural modernization contributed to people's willingness to adopt new strategies when faced with new realities, but intuitively the point of departure would seem to have been incipient mortality transformation.

In this way, writing mortality into the equation of demographic change was both a commonsense deduction from observed data and a convenient point of departure into the entire realm of the history of human rationality. Population growth rates before the nineteenth century were generally low, normally below 0.5 per cent per year, and whatever excess in fertility there might have been was very small. Within that context, no matter how much society might modernize, how literate women might be or what faith a person might have, it made little sense to curtail a completed family size which prevailing demographic realities limited anyway. The one variable which might alter this reproductive equation (strategy) was mortality, which could and did vary independently of family strategies and couples' choices. Basically the demographic transition, especially in its early stages, is difficult to imagine without some sort of change in mortality.

Yet results from empirical studies have often been ambiguous and contradictory on this matter. Chief among them were the largely disappointing conclusions coming from the Princeton Fertility Project, which led Francine van de Walle in her article summarizing the role of infant mortality in the demographic transition to assert, 'At the end of this quest, we cannot report that the historical evidence confirms that the declines of infant mortality led to the decline of fertility.'[5] Indeed, the results from those studies included in the project were mixed at very best. In Germany, Knodel found that marital fertility declined earlier than infant mortality in half of all German administrative areas, and otherwise mortality levels were only very modestly related to fertility in most contexts.[6] For Belgium, Lesthaeghe reached much the same conclusion: 'It is difficult to argue that the drop in infant mortality was among the factors that contributed to the start of the marital fertility transition in Belgium.'[7] Similar realities seem to have held for Italy and Portugal.[8] Even though other studies based on aggregate data sets have attributed a much more important role to mortality in the fertility transition, up until

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recently the general impression that it is a singularly unpromising variable was widespread among many scholars.[9]

What is the problem? While there may be some debate as to whether mortality decline was the initial and most significant spark to the onset of fertility control within marriage, it is difficult to imagine any prolonged fertility decline without some similar trend in mortality. The disconcerting results from the Princeton Project, the only large-scale research project which has dealt the subject of fertility–mortality interactions in the course of demographic modernization, stem in part from the perspective used to tackle the issue and from the choice of mortality indicator. The relationship between mortality and fertility over the demographic transition is a dynamic one by definition, one which is characterized to a large extent by the timing and intensity of change. It is a relationship of lags and leads. Even though the implications of mortality–fertility interactions are visible at a societal level, basically they are played out within families: their surviving children, their survival strategies, their reproductive choices. Capturing this acceptably well by using cross-sectional data aggregated by provinces is a quixotic quest, and can never yield more than very crude results. Further uncertainty was added by the decision to use infant mortality, itself only one part of overall childhood survival, as the sole mortality indicator. Family reconstitution studies offer a much more adequate vantage point for some aspects of this issue and, whenever they have been used to examine these interactions, the results have given support to the importance of child-survival for fertility. Unfortunately, studies addressing these issues are not numerous, and very few of them touch on the dynamics of demographic transition.[10] Until widespread use is made of family reconstitution and other techniques of family-level analysis applied to European data between 1860 and 1940, our understanding of the contexts of fertility decline will never be more than superficial and simplifying approaches to a very complex issue.[11]

In this article, I will present a dynamic context for the analysis of mortality and fertility in the course of the demographic transition, based on the experience of a sizeable sample of rural parishes taken from the central part of Spain. While this perspective cannot afford the wealth of information that one based on, say, family reconstitution does, nor can it claim to represent the experience of society as a whole as cross-sectional analysis does, it affords a glimpse of this process which neither of the others can. For one thing, the analysis can be couched in purely dynamic terms, taking into account the timing and intensity of changes in both mortality and fertility, and evaluating the ways in which they affect each other. Secondly, by taking the data from a specific, though large, sample of villages, we can see local variations relatively uncluttered with much of

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the cultural, social and economic heterogeneity which dogs studies based on very large levels of aggregation. This perspective will shed new light on the way in which we understand the process of demographic change, and will provide very strong evidence for the central role played by mortality change during the process of the demographic transition.

MORTALITY AND FERTILITY: AN INTERPRETATIVE FRAMEWORK

This analysis is based on some fundamental conceptions of the way in which mortality and fertility interrelate and on the heterogeneity of mortality decline itself. None of these ideas is startling or new, but they will affect the way in which we can expect our variables to perform.

Overall mortality should only have very modest and indirect effects on fertility within marriage unless it is the spouse who dies. It is during the first years of life that the influence of mortality will be strongest, as it constrains the number of children surviving past the initial dangerous years of their existence. Mortality during childhood could affect fertility in three ways, one of them directly and the other two mediated by rational choice. In the first place, the death of a child during the first months of life by definition could not help but have a fertility-boosting effect, as the death of infants halted breastfeeding and left mothers without its fecundity-inhibiting effects.[12] This type of link can only affect mortality during lactation, and implications for fertility should be visible fairly shortly thereafter. It is unclear, however, just how strong this effect can be and how long it takes to materialize.

The 'replacement effect' is considered a second way in which families compensate for higher than expected mortality.[13] Here there is a conscious effort to replace a child lost in order to meet the desired number of surviving children. It is very unlikely that this effect was ever very strong in 'natural fertility' regimes, mainly because high and variable levels of childhood mortality left little margin for choice. As mortality diminished and became more stable and as fertility began to be controlled, however, this process may well have become increasingly significant.[14] Any replacement effects should be visible during the years following a child's death.

The third way in which childhood mortality affects fertility is also mediated by rational choice. It is related to the replacement effect, but normally works in the opposite direction. Parents have always had certain reproductive goals in mind, and were well aware both when they had too many or too few surviving children.[15] In pre-transitional mortality regimes, their ability or willingness to act upon reproductive outcomes was severely limited. As mortality during childhood began to experience

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significant declines, however, increasing numbers of parents began having 'too many' surviving children. By the time they realized this, their only directly reproductive options were to limit the number of pregnancies they were to have or to stop them altogether. It was this sort of awareness that informed the parity-specific types of fertility limitation which seem to have been so widespread during the demographic transition in Europe.[16] By definition, however, this effect only took place after an appreciable lag. In other words, a mother could not be sure she was going to have too many surviving children until a certain number of them (say, three or four) had survived early childhood. By then the mother was necessarily nearing the final stage of her reproductive period, and any fertility control ended up taking the form of parity-specific limitation or stopping.

The experience of mortality during childhood was not uniform, responded to different risk environments and interacted with fertility in different ways. One of the most important oversights of the Princeton Project was to use infant mortality as the only index of childhood mortality experiences. The risk of death at other ages, most notably between one and three years of age, was often quite high. Judging from the mortality experience of a number of European countries during the latter part of the nineteenth century, mortality between one and four years of age represented between 22 and 45 per cent of all deaths of children under five years of age.[17] At the very least, using only deaths of children 0 years of age is dangerous because it only includes part of the picture.[18] Moreover, there is also strong reason to believe that the onset and the pace of mortality decline was often quite different for infants as opposed to child mortality.[19] By and large, it seems tht child mortality was the first to change, followed by infant mortality often as much as two or even three decades later.[20] Once the decline had set in, everywhere child mortality declined faster, leading to an increasingly younger age structure of childhood death over the course of the epidemiological transition.

The mortality of children (one to four years old) was not necessarily the product of the same disease environment as that of infants. The major difference here is based on the fact that infants often died of congenital or birth-related defects and in most cases lactation limited their contact with food-related diseases. Older children and infants above about six months of age, on the other hand, were more susceptible to diseases such as measles or smallpox, as well as to diseases of the digestive tract related to the entire process of teething and weaning. Reductions in infant mortality were largely related to infant feeding and care practices, as well as to the nutritional status of both the mother and the child. This was not necessarily the case with older children where environment-related problems were relatively more important. The mixture of factors leading

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to the decline of infant mortality was similar but not the same as that relating to child mortality, and there is no reason to expect declines to be identical in timing or in intensity. While much must still be done to understand differing risk environments during the course of childhood, the historical record gives abundant indications that the process of decline itself was not uniform.

With these points in mind, I would like to propose a framework for interpreting the role of mortality change in the transition of fertility. It is a synergistic and sequential framework in which mortality decline leads to fertility decline, and this in turn stimulates more accelerated mortality decline. The model has a specific point of departure – the early decline of childhood mortality – though this is related in turn to an entire set of social factors which also condition people's willingness to adopt fertility-control practices. Nevertheless, the role of mortality reduction is clear throghout. The framework is based on the hypothesis that a sequence of dynamics takes place, though the exact order of events may change in any given circumstance (see Figure 1). It should be noted that, in this explanatory framework, mortality–fertility interactions have been isolated so as to make it possible to pinpoint more clearly how they materialized. Ultimately, however, mortality was not and could not be the only factor for fertility limitation, though it may well have been the initial trigger which set the entire process of fertility reduction in motion.

Within this framework, mortality decline itself is the product of both exogenous and endogenous forces. In many ways, the social, economic and cultural context serves as a background variable conditioning the timing and intensity of changes in mortality as well as the willingness of families to adopt forms of fertility reduction in the face of increased numbers of surviving children. On the other hand, mortality reduction may also be related to factors which have little or nothing to do with people's attitudes towards children. The prevailing disease environment or the nutritional status of parents and children are examples of these.

The mortality of children between one and four years of age was the first childhood mortality indicator to decline, somewhere between five and thirty years before infant mortality. This reduction in mortality tended to increase the numbers of children surviving past the ages of greatest risk (say, the first five years of life), though its importance with respect to family strategies ended up depending on just how high mortality in those age groups tended to be. In a southern European context, where child mortality (${}_4q_1$) was as high or higher than infant mortality (IMR), its importance would have been greater than in other areas where it was less severe.[21] Everywhere, however, the effects of declining child mortality must have been felt. Infant mortality reduction was itself the product of

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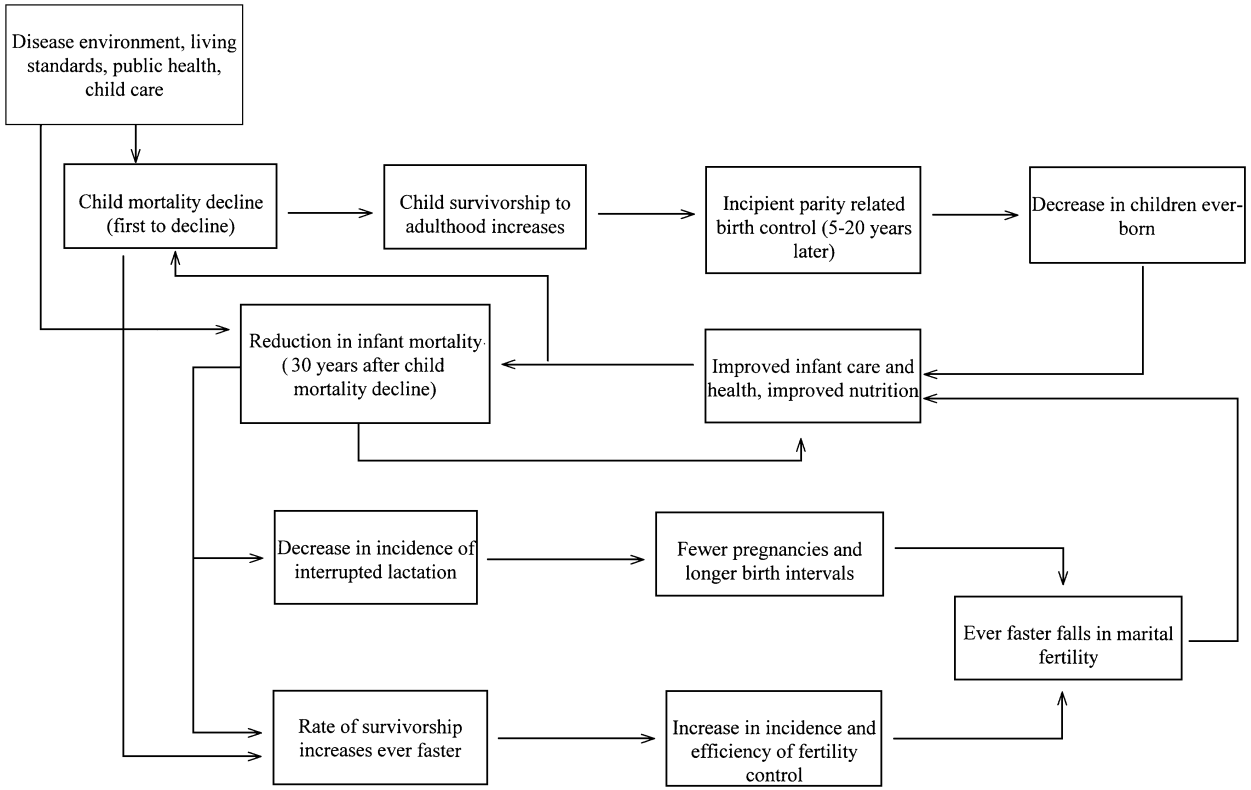

FIGURE 1. Mortality–fertility interactions during the demographic transition: a model of dynamic synergy.

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certain exogenous factors, as well as the general context of improved infant care resulting from the initial fertility decline. Improved infant survivorship had a dual effect on fertility. Like child mortality, it had a delayed effect on fertility as the chances of having children survive past five years of age grew at an ever-faster rate. In addition, infant mortality decline had a more immediate fertility-reducing effect because it tended to decrease the incidence of interrupted breastfeeding. From the standpoint of fertility, the immediate effect tended to increase birth intervals for mothers in all age groups, whereas the delayed effect would only have been visible among women of higher ages, at least in its initial stages.

Once overall levels of survivorship had risen sharply, families realized that they were going to have too many surviving children, certainly more than their parents had had. Families in this situation must have always existed. The difference is that previously families with favourable mortality experience were a minority, whereas once general mortality decline had set in they became the majority. Key to this entire framework is the fact that families were aware of the negative economic implications of having too many surviving children, both for living standards and for the viability of future intergenerational property transfers. Had migratory opportunities been limitless or had these potentially detrimental economic effects not existed, it is highly unlikely that families would have ever taken steps to limit their fertility in the face of increasing number of surviving children. While this may have happened temporarily in many countries of the developint world during the central decades of the present century,[22] it hardly seems to have been a significant factor in most of Europe during the demographic transition. Thus, fertility reduction can be viewed in many ways as a compensatory mechanism in the face of the changing demographic and economic circumstances perceived by families.

It is unclear how much childhood mortality had to decline before families realized its implications and acted upon them. It is quite possible that in the very early stages of the mortality transition initial, often halting, reductions in mortality were not perceived by families as indications of the start of long-term trends but rather as cyclical changes, and thus they may well have had fairly muted effects on fertility. Once the decline gathered strength, however, the relationship between both variables should become much stronger. Moreover, if mortality gains were concentrated in certain families, as they likely were, the net effects for them would be far greater. Families in this situation must have been the pioneers in implementing certain fertility-control practices.[23] These were probably very inefficient at first, mainly *coitus interruptus*, but they ended up reducing their fertility. It was a limitation which was partly-specific and took place in higher age groups. It is quite possible that this sort of

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inefficient fertility limitation had always existed among families with favourable mortality experience, though the mortality transition increased their numbers dramatically.

Declining fertility in turn became a powerful force reducing mortality still further because child spacing became greater, therefore affording parents more time with each child. In this sense infant mortality decline was probably more significant because it also had a direct effect on child spacing and this in turn was a powerful factor for improved childhood survival.[24] The importance of fertility decline for further mortality reduction should not be underestimated, especially after the initial onset of decline in both indicators.

EXPLORING HISTORICAL REALITY

What was presented in the preceding section is a very schematic portrayal of mortality–fertility synergism during the course of the demographic transition. In our attempt to focus on the relevant role of mortality for fertility decline, many other important factors have been left out entirely, or their influence has been minimized. The framework suggests that there should be certain observable lags and leads in the process, and that the rate of decline of both fertility and mortality should increase with the passage of time. It is not difficult to ascertain the empirical validity of these propositions by looking at the historical record. This task will occupy much of the rest of this article.

Since the framework implies a type of sequential causality, we have chosen to use time series in order to illustrate the patterns most clearly. Data have been taken from a large sample of villages in central Spain over the course of the nineteenth and twentieth centuries.[25] All childhood mortality indicators used in this paper have been calculated directly from the births and deaths by age, and thus are subject to constraints imposed by migration at young ages. For marital fertility we have used the index I_g, designed within the context of the Princeton Fertility Project. The denominator for this index has been derived from the census-based age and marital status of the female population in 1887, 1900, 1910, 1920, 1930, 1940, 1950 and 1960.[26] Legitimate births have been estimated by applying legitimacy rates in rural areas of the provinces of Madrid and Toledo after 1900 to the total births, taken from local sources.[27] Whenever these indicators have been contrasted with independently generated estimates of childhood mortality or fertility, the results have been fully satisfactory, though improving registration of births and deaths may explain at least some of the trends visible during the central decades of the nineteenth century. Nevertheless, we feel that these data represent the

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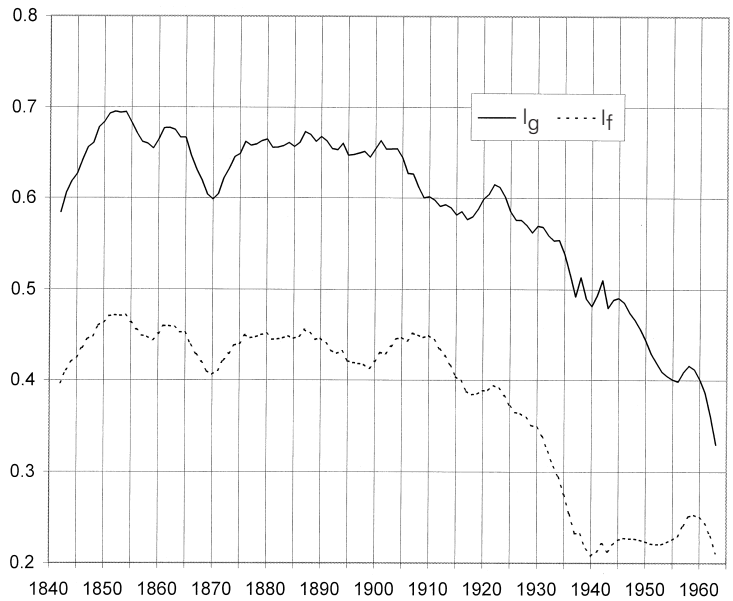

FIGURE 2. I_g and I_f in a sample of rural New Castile, 1840–1965 (five-year moving average).

actual trends in fertility and mortality in a significant area of Spain, both in terms of levels and of timing and intensity of trend changes. Five-year moving averages have been used so as to portray relatively short-term changes accurately. At the onset of the demographic transition, the central areas of the country were afflicted by low life expectancy (e_0 = 29–32), high infant mortality (well above 200 per thousand) and child mortality which was equally as high or higher. Pre-transitional marital fertility was only moderately high (I_g = 0.600–0.700), though overall fertility (I_f) was fairly high thanks largely to very low levels of permanent celibacy (often below 4–5 per cent).

Spanish data have been used because very little is known of fertility and mortality trends in Spain before the twentieth century. Spain, however, is only a case in point. The framework presented here should be applicable to any other European context. The basic patterns emerging should be similar, with variations being due to certain contextual variables such as the levels of mortality, the ratio of child to infant mortality, the standard of living of the population, or the pace of implementation of public health reforms. The overall picture, however, should remain substantially the same.

Fertility trends in the rural areas of central Spain can be seen in Figure 2 where I_g and I_f have been plotted from the central decades of the

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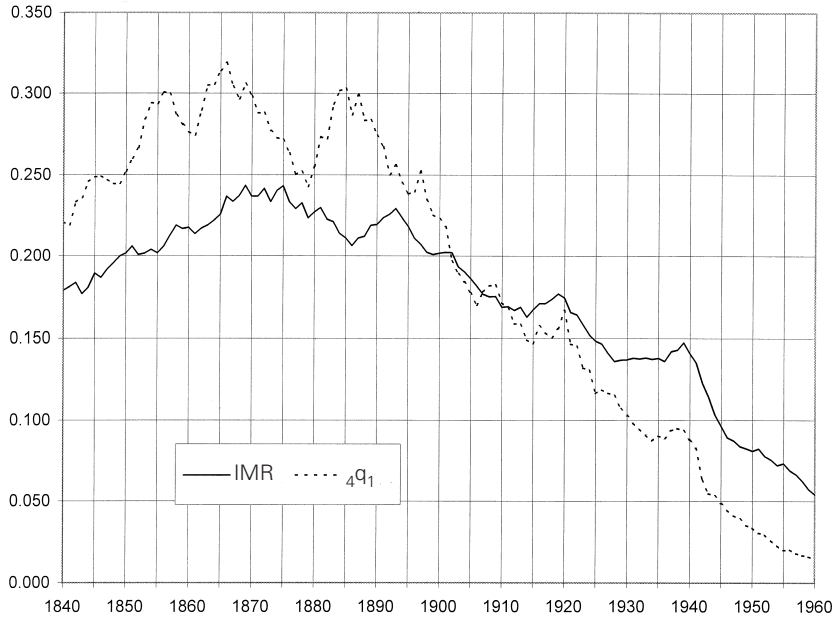

FIGURE 3. Infant (IMR) and child ($_4q_1$) mortality in a sample of rural New Castile, 1840–1960 (five-year moving average).

nineteenth century. After a period of fairly steady fertility, marital fertility peaks towards the end of the 1880s and begins a slow but almost unbroken descent after that moment. Ten years after the decline set in, fertility was only down by 4–5 per cent. Much more intense and consistent decline started just after the turn of the century, and by 1915 marital fertility stood 10 per cent below levels holding in 1905. Even though this region may not have qualified for the arbitrary Princeton definition of the 'onset of fertility decline' until the decade of the 1920s or the 1930s, by that time it was already 15–17 per cent beneath its peak and fertility decline had been unquestionable for several decades. General fertility (I_f) did not begin its trend downward until after 1910, a decade after marital fertility.

Figure 3 contains two mortality indicators for early childhood, and gives very strong proof that the reductions in child mortality ($_4q_1$) preceded those in infant mortality by about ten years.[28] After increases in both indicators during the central decades of the nineteenth century, child mortality began to decline during the mid 1860s, and infant mortality followed suit around 1875. Mortality decline was not uninterrupted, as a series of epidemics reversed the downward trend of child mortality in the mid 1880s, and of infant mortality in the mid 1890s. Afterwards rates of decline become much greater and were only interrupted during the

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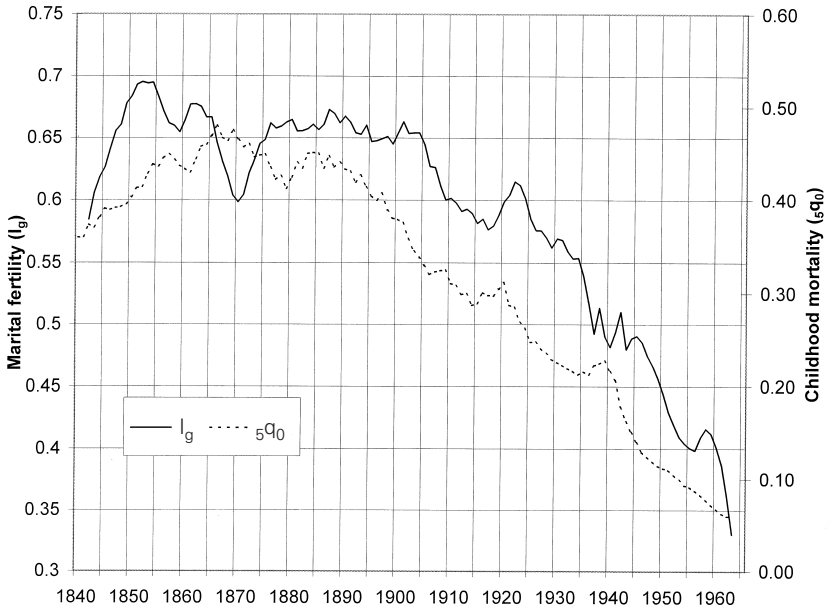

FIGURE 4. Marital fertility (I_g) and childhood mortality ($_5q_0$) in a sample from rural New Catile, 1840–1965 (five-year moving average).

influenza epidemic of 1918–1920 and then again during the Civil War and its immediate aftermath. Child mortality fell by nearly 23 per cent during the first ten years (1885–1895), and infant mortality did so by over 15 per cent (1893–1903). In the process of mortality transition, older children experienced gains which were relatively greater than those of infants, as $_4q_1$, which in pre-transitional contexts had been significantly higher than IMR, dropped beneath infant mortality by the beginning of the twentieth century.

Taken together, Figures 2 and 3 give confirmation to a number of ideas raised in the interpretative framework outlined earlier. These can be summarized in the following points.

First, the key decline in child mortality preceded that of marital fertility by around 15 years, whereas infant mortality did so at a shorter lag of about 7–8 years. In fact, a dual pattern can be seen. The first and subsequently interrupted decline in child mortality preceded the first timid declines in marital fertility by 20 years, as did infant mortality by about 10–12 years. The gap was shorter for the second and more intense mortality decline which preceded uninterrupted fertility reduction by between 7 and 16 years, depending on the indicator. If we use childhood mortality ($_5q_0$) as our mortality indicator (see Figure 4), it is clear that the

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decline in mortality occurred 20 years before the initial decline of marital fertility, which only reacted when mortality was already 25 per cent off the peak reached in the mid 1880s.[29]

Secondly, fertility decline tended to accelerate about 5–10 years after both mortality indicators were in steady decline. In the period following the incipient decline of child mortality and when infant mortality was just beginning to fall, marital fertility fell by a very modest amount. In the following decade (1905–1915) marital fertility began an extremely rapid fall, following a decade (1895–1905) of very intense mortality reduction. Between 1905 and 1920 mortality decline slowed, with the result that between 1915 and 1925 marital fertility was practically stable or even increased. In the following period (1920–1930), mortality declined much more quickly (25 per cent in both cases), and as a result fertility reduction also accelerated. While the relationship is not perfect, it does indicate that the rate of decline of fertility seems to have been sensitive to the decline in mortality indicators during the period immediately before. It also suggests that the very timid incipient declines in mortality had little effect on fertility. Only when mortality was well off its peak, and immersed in what was to be irreversible transition, were its effects on fertility clearly evident.

Thirdly, throughout much of the period, swings in mortality seem to be followed by similar swings in marital fertility, though the lag was normally somewhat greater with child mortality than it was with IMRs. This apparent relationship is visible not only during the initial period of the demographic transition, but it also seems to have increased in the course of fertility and mortality reductions. These similarities suggest that the links between mortality and fertility were long-standing, and may have characterized both pre-transitional and transitional populations.

Finally, there is some indication that the gap between mortality and fertility trends began to narrow during the course of the twentieth century, when both fertility and mortality are well off their high points. This may indicate the existence of a type of replacement effect in which periods of temporarily increasing mortality – especially during the twentieth century due to the influenza epidemic of 1918–1920 and the Civil War and its immediate aftermath – were followed at very short intervals (2–5 years), by temporary upturns in marital fertility.

These last two points can be seen more clearly if the time series are detrended in order to remove the overall downward trend of all variables. Here we have used two different detrending techniques because of the different nature of the links we are attempting to see. One of them is the direct, almost structural link between infant mortality and fertility through interrupted breastfeeding, and the other is the more distant and

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TABLE 1

Short-term interactions between infant mortality (IMR) and marital fertility (I_g): lagged correlation coefficients between detrended indicators of infant mortality and marital fertility

1851–1953		*1851–1899*		*1900–1953*	
Lag	*IMR with I_g*	*Lag*	*IMR with I_g*	*Lag*	*IMR with I_g*
0	−0.164	0	−0.202	0	−0.152
1	0.058	1	−0.126	1	0.143
2	0.037	2	0.055	2	0.031
3	0.168	3	−0.034	3	0.266
4	0.142	4	0.214	4	0.117
5	−0.046	5	−0.042	5	−0.049

Note: All series detrended by dividing yearly values by an 11-year moving average.

indirect link between both variables as mediated by people's choice. The first of these interactions should be visible fairly close to the death of a child and the other will materialize several years later. For the first case, yearly values of I_g and of IMRs have been divided by eleven-year moving averages, and for the second the five-year moving average of each indicator was divided by its twenty-five-year moving average. The first method will yield a clear picture of year-to-year changes and seems appropriate for the direct, almost immediate link, while the second technique will give us an approximate idea of medium-term trend changes in both indicators and seems to be more appropriate for seeing the longer-term more diffused link. In both cases, the long-run trends have been removed, preserving the shorter trends which are the key to our discussion. In all cases, the mean of the detrended series is very close to 1. Lagged bivariate correlation coefficients between I_g and the fluctuations in mortality in earlier years have been used as a simple, straightforward analytical technique. The results of both exercises are contained in Tables 1 and 2, and tend to support many of the points raised earlier.

Table 1 suggests that increases in infant mortality led to similar jumps in marital fertility, though over a period of five years this relationship was not uniform. As expected, mortality and fertility were inversely correlated at lag 0, though the correlation becomes modestly positive over the next four lags. Not surprisingly, given the detrending techniques, the correlation coefficients are not high but are consistently positive, with the highest values coming at lags 3 and 4. It is interesting to note that when the correlation analysis is run on nineteenth-century data, a good part of the link between infant mortality and fertility disappears, with the

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TABLE 2

Medium-term mortality–fertility interactions between childhood mortality and marital fertility (I_g): lagged correlation coefficients between detrended indicators of infant and child mortality and marital fertility

Detrended series (1851–1953)			*Detrended series (1851–1899)*			*Detrended series (1900–1953)*		
Lag	*IMR with I_g*	*${}_4q_1$ with I_g*	*Lag*	*IMR with I_g*	*${}_4q_1$ with I_g*	*Lag*	*IMR with I_g*	*${}_4q_1$ with I_g*
5	0.333	0.202	5	0.025	−0.261	5	0.456	0.442
6	0.374	0.275	6	0.143	−0.134	6	0.472	0.493
7	0.419	0.325	7	0.260	0.007	7	0.495	0.501
8	0.389	0.278	8	0.291	0.081	8	0.442	0.393
9	0.321	0.228	9	0.303	0.159	9	0.341	0.275
10	0.215	0.182	10	0.266	0.158	10	0.201	0.204
11	0.120	0.074	11	0.208	0.099	11	0.086	0.068
12	0.015	−0.062	12	0.203	−0.010	12	−0.054	−0.085

Note: All series detrended by dividing 5-year moving averages by 25-year moving average.

exception of births taking place four years after a change in mortality. In the twentieth century, however, correlations tend to be higher, due perhaps to what appears to be the incipient existence of the replacement effect and perhaps the incipient willingness of families to use birth-spacing techniques when faced with increases in child survivorship.

Medium-term mortality–fertility interactions can be seen in Table 2. Assessing the period as a whole, noticeable links appear between mortality trends and fertility several years later. These links are greater with infant mortality than they are with child mortality, probably because the bulk of children dying were infants and their relative importance in childhood mortality increased over the course of the demographic transition. The closest tie appears to have materialized seven to eight years after a change in mortality conditions. When the analysis is run on nineteenth-century data, the pattern is similar in very general terms, though with notable exceptions. Here the gap between mortality and fertility is strongest after nine to ten years, the coefficients are generally lower than for the period as a whole, and links between ${}_4q_1$ and I_g at shorter lags disappear or become negative. During the twentieth century, however, this all changes. The mortality–fertility links are much stronger, and the effect is much less delayed.

These results seem to suggest that, in the course of the demographic transition, mortality–fertility interactions became stronger and more

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immediate. As mortality declined, it also became less volatile and easier to predict; and as fertility declined and couples' reproductive goals could be met with fewer children, families had more room to make decisions. Since, however, the pace of mortality decline tended to speed up, these family reproductive decisions tended to be in the direction of increasing fertility limitations. In this way, people became increasingly aware of their reproductive goals much earlier on in their marriages and were willing to adopt spacing as well as stopping techniques of fertility control in order to meet these goals in the face of ever-decreasing mortality.

DISCUSSION

The Castilian data set has afforded a very straightforward example of a number of the points raised in the first two sections of this article. Mortality decline was not uniform either in the timing of its onset or in the intensity of the decline, as child mortality declined earlier and faster than infant mortality. Initial declines in marital fertility came eight to nine years after those in infant mortality but fifteen to twenty years after child mortality began its historic descent. When the entire childhood mortality experience is included, the gap between declines in fertility and mortality is actually closer to twenty years. Throughout the demographic transition, moreover, there seems to have been an ongoing sensitivity of marital fertility to childhood mortality, though the lag was invariably different for each mortality indicator. An explanation of this pattern was advanced within the context of the analytical framework proposed and was based on parents' lagged awareness of potentially having too many surviving offspring, together with the almost immediate effect of infant death on birth intervals and child spacing.

The fairly prolonged gap in the initial trend changes of mortality and fertility indicators seems to suggest that the first reductions in mortality were not in themselves sufficient to influence women's reproductive behaviour. It was only when these changes continued and when mortality was well off its peak that fertility control first made its appearance. In other words, over a period of ten years or more women must have been able to witness the effect of continued and cumulative mortality decline. By the time fertility actually began to fall during the first years of the twentieth century, overall childhood mortality (${}_5q_0$) was 25 per cent below its peak of the late 1880s.[30] Moreover, after fifteen years of uninterrupted decline, it was clear that this was not just a short-term trend but rather a substantial change with far-reaching effects. It was then, and only then, that fertility first underwent significant reductions. A woman, say, 30 years old in 1905 had spent her entire reproductive period in a situation

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of stable fertility and sharply declining mortality, with the result that the number of children surviving past early childhood was substantially higher than those surviving when she was a child. The implications of this for the well-being of her family were clear.

Techniques of fertility limitation within marriage had always been known to European populations, and had been selectively used in periods of economic and/or political crisis, during epidemics and, in some contexts, even during Lent and Advent in the past.[31] The difference now was that the use of these often inefficient techniques was no longer intermittent as it had been in the past. The need to approximate reproductive goals to the reality imposed by changing childhood mortality ended up being recognized generally, especially as the rate of mortality decline increased.

The way in which this materialized was within the context of family strategies and family decision-making processes. This is one of the reasons why studies of the demographic transition based on large aggregates have traditionally been so disappointing when addressing the mechanisms of change. Ideally family reconstitution, especially if it included socio-economic and cultural variables, would be the method of choice if we were to attempt to pinpoint the covariates of fertility control, though this technique would still make it very difficult to understand the timeliness of changing relations which was so essential to the entire process. The type of data used in this article also have serious limitations, but they help us understand that the entire process of transition was to a very great extent one of lags and leads, especially insofar as the childhood mortality–fertility links are concerned.

While the data used here are Spanish in origin, there is no reason at all to think that the basic process did not hold within other very different contexts. The intensity of the relationship between infant or child mortality and fertility might vary due both to the levels involved and to the timing and pace of decline. In those societies in which child mortality was relatively low, its decline would have a much smaller and more delayed impact on fertility than where it was high. Infant mortality would everywhere be more important due to the dual nature of its effect on fertility and because the majority of children dying were invariably infants. This preeminence, though, would have been even more visible when infant to child mortality ratios were relatively high, as they were in northern Europe. Everywhere, however, declines in both types of childhood mortality would be important.

It is difficult to find the requisite data from other contexts to contrast these hypotheses; infant mortality and marital fertility indicators can be gleaned from the historical records of some European countries, though

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child mortality levels are more problematic. Where this has been possible, the data suggest that the basic pattern described here remains true. In England and Wales, the initial fertility decline dates approximately from 1880 and child mortality (${}_4q_1$) begins its uninterrupted fall during the decade of the 1860s.[32] Infant mortality evidences a more complex trend, as initial decline sets in during the decade of the 1870s or earlier, though there is a rebound between 1880 and 1895. Recent studies, however, suggest that this increase was due to the weight of the urban population where mortality was high until the end of the century, and that in rural areas the decline started in the 1870s and continued uninterrupted.[33] In either case, some type of mortality decline preceded fertility decline in England by as much as fifteen years. In Sweden, the lags are different but the pattern is the same. Marital fertility began to decline around 1880, infant mortality around 1870, early child mortality (1–3 years) around 1865 and later child mortality (4–5 years) near 1860.[34] Apart from these two examples, very little else has appeared on this subject. Knodel's study of German village populations offers some support, though the nature of his data base makes it almost impossible to pinpoint trend changes with any sort of reasonable accuracy.[35] Generating a reliable picture of the pace of mortality transition during childhood in Europe will not be possible until future research brings to light the patterns holding in other contexts.[36]

What our data do not say is just how this process spread to the entire society. We have postulated that the first families to control their fertility were the ones with the most favourable mortality experience. This had likely always been the case though, with the possible exception of some areas of France and perhaps Switzerland, these families were never numerous enough to end up setting an example for the rest of society or to make aggregate fertility indicators decline significantly. During the second half of the nineteenth century, however, mortality during childhood began to decline everywhere, due in part to the remission of certain childhood diseases, to improved nutrition and to the incipient awareness of rudimentary methods of child care and hygiene among mothers. While different areas and social groups experienced faster declines than others, the phenomenon was widespread. This last aspect was particularly important because such rudimentary methods of child care could be learned easily and used efficiently. We do not know who the first families were who knowingly began to apply these techniques, but they were likely the ones where mortality declined first and most, and consequently the first to control their fertility. It was only a matter of time, and not much at that, before the success of the families with better child care practices, higher child survival rates and fewer childbirths became an

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example for other members of society. In other words, the process whereby mortality and fertility declined ended up being largely one of diffusion from the pioneers to the rest of local society.[37] Tracking those leading families and the role they played in their communities would seem to be essential for understanding the entire process of transition.

It is, however, important to note in this context that many of the variables informing incipient fertility control were probably the same ones working to depress mortality, or were working through the intermediate variable of mortality. In other words, fertility transition came from the same context as mortality transition. Families with harsher mortality experiences were probably always the ones least likely to do anything at all to limit their fertility. This was probably as true during the initial stages of the transition as it was further along when mortality and fertility had both declined substantially. There are of course a number of exogenous variables at work here, especially insofar as the decine of mortality is concerned. Nevertheless it is unquestionable that the context of mortality and fertility transition was largely the same, and that it spread throughout society thanks to the diffusion of the behaviour patterns of those who had initially practiced modern methods of child care and fertility control to their own benefit and that of their society.

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ENDNOTES

1 Frank W. Notestein, 'Population: the long view', in T. Schultz ed, *Food for the world* (Chicago, 1945).

2 Abdel R. Omran, 'The epidemiologic transition: a theory of the epidemiology of population change', *Milbank Memorial Fund Quarterly* **49**, 4, pt 1 (1971), 509–38.

3 On this point, see Jean-Claude Chesnais, *La transition démographique: étapes, formes, implications économiques* (Paris, 1986), 141–54.

4 Slave populations would be the classic example of populations with low fertility and high mortality.

5 Francine van de Walle, 'Infant mortality and the European demographic transition', in A. Coale and S. C. Watkins eds., *The decline of fertility in Europe* (Princeton, 1986), 201–33, esp. p. 233. See also Poul C. Matthiessen and James C. McCann, 'The role of

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mortality in the European fertility transition: aggregate-level relations', in S. Preston ed., *The effects of infant and child mortality on fertility* (New York, 1978), 47–68, esp. pp. 62–7).

6 Knodel affirmed, 'It appears that the usual description of the demographic transition which postulates a prior decline in mortality (particularly infant and child mortality) as an initiating cause of fertility decline does not fit the facts in Germany'. See John E. Knodel, *The decline of fertility in Germany, 1871–1939* (Princeton, 1974), esp. pp. 167–85.

7 Ron J. Lesthaeghe, *The decline of Belgian fertility, 1800–1970* (Princeton, 1977), 171–6.

8 See Massimo Livi-Bacci, *A century of Portuguese fertility* (Princeton, 1971), 121–4, and Massimo Livi-Bacci, *A history of Italian fertility during the last two centuries* (Princeton, 1977), 205–13. Much the same conclusion is reached in other studies related with this project. See, for example, the case of England in Michael S. Teitelbaum, *The British fertility decline: demographic transition in the crucible of the Industrial Revolution* (Princeton, 1984).

9 Examples of this pattern can be found in the provinces of rural Spain between 1887 and 1930 and in the *Kreise* of Prussia between 1875 and 1910. See David S. Reher and Pedro Luis Iriso-Napal, 'Marital fertility and its determinants in rural and in urban Spain, 1887–1930', *Population Studies* **43** (1989), 405–27, esp. pp. 411–18, and Patrick R. Galloway, Eugene A. Hammel and Ronald D. Lee, 'Fertility decline in Prussia, 1875–1910: a pooled cross-section time series analysis', *Population Studies* **48** (1994), 135–58, esp. pp. 151–7. See also Ronald D. Lee, Patrick R. Galloway and Eugene A. Hammel, 'Fertility decline in Prussia: estimating influences on supply, demand and degree of control', *Demography* **31**, 2 (1994), 339–65. In a recent (1998) paper, Galloway, Lee and Hammel have offered the strongest proof to date that, at least in the case of Prussia, infant mortality appears to have played a significant role both for fertility levels and especially for fertility decline; see Patrick R. Galloway, Ronald D. Lee and Eugene A. Hammel, 'Secular marital fertility decline and infant mortality: a reassessment of the European macro evidence', in M. Montgomery and B. Cohen eds., *From birth to death: mortality decline and reproductive change* (Washington DC, 1998, 182–225). In a recent article, Garrett and Reid refer to mortality as an 'enabling' factor for the decline of fertility in England, though in the long run they are reluctant to attribute to it a causal role in fertility decline; see Eilidh Garrett and Alice Reid, 'Thinking of England and taking care: family building strategies and infant mortality in England and Wales, 1891–1911', *International Journal of Population Geography* **1** (1995), 69–102, esp. pp. 98–9. In a recent study based on local data from northern Spain, Sánchez found that mortality during the first fifteen years of life had a significant effect on marital fertility decline. See Jésus Javier Sánchez, 'Mortality and the fertility transition in rural Navarre (Spain)', *Genus* **14**, 1–2 (1998), 57–75, esp. pp. 68–73. For a contrary view with respect to the experience of the United States, see the recent article by Michael R. Haines, 'The relationship between infant and child mortality and fertility: some historical and contemporary evidence for the United States', in Mark R. Montgomery and Barney Cohen eds., *From deaths to births: mortality decline and reproductive hange* (Washington, DC, 1998), 227–53.

10 The most complete of these studies is that of John Knodel on 14 German village populations. See John E. Knodel, *Demographic behavior in the past: a study of fourteen German village populations in the eighteenth and nineteenth centuries* (Cambridge, 1988), esp. pp. 393–441. Unfortunately, Knodel's study ends towards the latter part of the nineteenth century, just when the marital-fertility transition was beginning to get under way in most regions of Germany. Elsewhere, in a literature review of existing family

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reconstitution studies Knodel found that higher mortality led to higher parity progression ratios, later cessation of childbearing and a higher mean number of children ever born. See John Knodel, 'European populations in the past: family-level relations', in S. Preston ed., *The effects of infant and child mortality on fertility* (New York, 1978), 21–45. The French and Swiss studies reviewed, however, referred by and large to the seventeenth and eighteenth centuries.

11 For an assessment of some of the difficulties affecting the study of this issue from the standpoint of family reconstitution, see Galloway, Lee and Hammel, 'Secular marital fertility decline', 195–6.

12 This basically acts by prolonging postpartum amenorrhea. For more on this, see Jeroen K. van Ginneken, 'The impact of prolonged breastfeeding on birth intervals and on postpartum amenorrhoea', in W. H. Mosley ed., *Nutrition and human reproduction* (New York, 1978), 179–95; Robert G. Potter, 'Changes in natural fertility and contraceptive equivalents', *Social Forces* **51**, 1 (1975), 36–51; John E. Knodel, 'Breast-feeding and population growth', *Science* **198** (1977), 1111–15; and especially Knodel, *Demographic behavior in the past*, 395–405.

13 On this point, see Jacques Vallin and Alain Lery, 'Estimating the increase in fertility consecutive to the death of a young child', in Preston ed., *The effects of infant and child mortality*, 69–90; and Samuel H. Preston, 'Introduction', ibid., 1–18.

14 See Preston, 'Introduction', 12–13.

15 There is no need to insist here on the value of children to their families both before and during the demographic transition. Families were aware of their importance in the fields, in caring for siblings, in perpetuating the family lineage and for care of their parents when they were elderly. Insuring an adequate number of surviving children was of the utmost importance for most families. The literature on this subject is abundant. For a recent review, see David S. Reher, 'Wasted investments: some economic implications of childhood mortality patterns', *Population Studies* **49**, 3 (1995), 519–36.

16 For more on the incidence of parity-related fertility limitation in pre-transitional and transitional populations, see Ansley J. Coale, 'The decline of fertility in Europe since the eighteenth century as a chapter in human demographic history', in Coale and Watkins eds., *The decline of fertility*, 1–30, esp. pp. 8–30. For doubts about the parity-specific model, at least when applied to Ireland, see Paul A. David and Warren C. Sanderson, 'Measuring marital fertility control with CPA', *Population Index* **54**, 4 (1988), 691–713. Garrett and Reid maintain that in England both spacing and stopping methods of family limitation were used; see Garrett and Reid, 'Thinking of England', 97–9.

17 See, for example, Reher, 'Wasted investments'.

18 See Mathiessen and McCann, 'The role of mortality', 52.

19 On this point, see for example van de Walle, 'Infant mortality', 206–8. Recent research suggests that the gap between declines in infant and early childhood mortality rates may have been as great as 30 years in England and Wales, though this gap may be distorted by the weight of urban populations. See Robert Woods, Naomi Williams and Chris Galley, 'Infant mortality in England – 1550–1950: problems in the identification of long-term trends and geographical and social variation', in C. Corsini and P. P. Viazzo eds., *The decline of infant mortality in Europe, 1800–1950: four national case studies* (Florence, 1993), 35–50, esp. pp. 37; and E. Anthony Wrigley, *Population and history* (New York, 1969), Fig. 5.3. In Sweden it was between 5 and 10 years. See Gustav Sundbärg, *Bevölkerungsstatistik Schwedens 1750–1900* (Stockholm, 1907; reprinted with preface and vocabulary in English in *Urval: Skriftserie Utgiven av Statistiska Centralbyrån* **3** (1970), esp. p. 131. In areas of northern and central Spain, on the other

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hand, the gap seems to have been closer to 10–20 years, depending on the context; see David S. Reher, Vicente Pérez-Moreda and Josep Bernabeu-Mestre, 'Assessing change in historical contexts: childhood mortality patterns in Spain during the demographic transition', in C. Corsini and P. P. Viazzo eds., *The decline of infant and child mortality: The European Experience*, *1750–1990* (The Hague, 1997), 35–56; and Mercedes Arbaiza, Ana Guerrero and Arantza Pareja, 'Mundo rural y mundo urbano en la transición de la mortalidad vizcaína', *Boletín de la Asociación de Demografía Histórica* **XIV**, 2 (1996), 19–55. Our research on central Spain has shown that mortality of one- and two-year-olds led the way, many years before declines for higher and lower ages.

20 Galloway, Lee and Hammel have argued that infant mortality is an adequate indicator because infant and child mortality levels tend to be highly correlated in given societies. See Galloway, Lee and Hammel, 'Secular marital fertility decline', 186–7. While this may be true in general terms, using only infant mortality makes it impossible to assess the effects on fertility of earlier and more rapid declines in mortality after infancy which, in some cases, may have been quite substantial.

21 In our study, infant mortality (IMR) refers to those infants dying before their first birthday and child mortality ($_4q_1$) is the probability of death between a child's first and fifth birthdays.

22 In many developing countries the initial declines in mortality were met with steady or even increasing fertility for two or even three decades before fertility too began to decline.

23 This would be especially true in contexts where out-migration was not a ready option.

24 For more on this issue, see Galloway, Lee and Hammel, 'Secular marital fertility decline'.

25 At present this sample includes time series from 42 villages, mainly from the provinces of Madrid and Toledo, which represent a population of just under 120,000 people in 1887 and 145,000 in 1930. Taken together they represent between 18 and 25 per cent of the rural populations of those provinces. These data have been taken from both parish and civil registers. This is the only way to glean deaths by exact age from the existing sources because official Spanish publications only include this variable beginning in 1900, and even then it is only possible to estimate yearly series of certain limited mortality indicators. For more on this data base, see Reher, Pérez-Moreda and Bernabeu-Mestre, 'Assessing change'.

26 Since marital status by age does not exist prior to the 1887 census, before that year the number of women and the number of married women in each age group has been estimated by projecting backwards the totals from 1887 in accord with the overall population growth rate of the villages contained in the sample. This implies equal and constant rates of growth for each age group and an unvarying marriage pattern.

27 For the years before 1900, legitimacy rates corresponding to that year have been used.

28 The mortality levels from this sample of villages are quite similar to those taken from official vital statistics corresponding to the rural areas of the provinces of Madrid and Toledo during the first decades of the twentieth century. On this point, see Reher, Pérez-Moreda and Bernabeu-Mestre, 'Assessing change'; Alberto Sanz-Gimeno, *La transición de la mortalidad infantil y juvenil en el Madrid rural*, *siglos XIX y XX* (unpublished doctoral dissertation, Universidad Complutense, Madrid, 1997); and Diego Ramiro-Fariñas, *La evolución de la mortalidad en la infancia en la España interior*, *1785–1960* (unpublished doctoral dissertation, Universidad Complutense, Madrid, 1998).

29 If we take the initial decline in childhood mortality as our point of reference (1865–1870), the first declines in marital fertility took place between 35 and 40 years

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after the downward turn in childhood mortality and after it had dropped by about 30 per cent.

30 Child mortality ($_4q_1$) was already almost 33 per cent below its peak of the mid 1880s and infant mortality was 14 per cent below the high point reached in the early 1890s.

31 An example of this last point is the Spanish town of Cuenca where between the late sixteenth and the mid nineteenth centuries a shortfall in conceptions, and most likely in coital frequency as well, of 21 per cent has been shown to have existed during the Lenten season. See David S. Reher, *Town and country in pre-industrial Spain: Cuenca, 1550–1870* (Cambridge, 1990), esp. pp. 103–5.

32 For English data, see Wrigley, *Population and history*, Figs 5.3 and 5.4, and Table 5.15; and Woods, Williams and Galley, 'Infant mortality in England', 37.

33 On this point, see Robert I. Woods, P. A. Watterson and J. H. Woodward, 'The causes of rapid infant mortality decline in England and Wales, 1861–1921', *Population Studies* **42** (1988), 343–66 and **43** (1989), 113–32, esp. pp. 356–62; and Woods, Williams and Galley, 'Infant mortality in England', 41–42); and C. H. Lee, 'Regional inequalities of infant mortality in Britain, 1861–1971: patterns and hypotheses', *Population Studies* **45**, 1 (1991), 5–65. See also Naomi Williams and Graham Mooney, 'Infant Mortality in an "Age of Great Cities": London and the English provincial cities compared, c. 1840–1910', *Continuity and Change* **9**, 2 (1994), 185–212; and Garrett and Reid, 'Thinking of England'.

34 See Sundbärg, *Bevölkerungsstatistik Schwedens*, 121, 131.

35 Knodel was forced to group his data by 25-year intervals. He found that the high points of infant and child mortality and of marital fertility all come during the 1850–1874 period, though child mortality had begun to decline early in the century and was characterized during the 1825–1875 period by unchanging levels. See Knodel, *Demographic behavior in the past*, 42–3, 250.

36 For examples of recent research on this subject, see Alain Bideau, Bertrand Desjardins and Héctor Pérez Brignoli eds., *Infant and child mortality in the past* (Oxford, 1997), and Corsini and Viazzo eds., *The decline of infant and child mortality*.

37 Eilidh Garrett and Alice Reid support the idea of social and spatial diffusion of the small-family-size ideal in transitional England. See Garrett and Reid, 'Thinking of England', 98–9.

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